
Top 30 Viva Questions and Answers (E-Book Management System)

◆ 1. What is the main objective of your project?

Answer:

The main objective is to create a web-based platform where users can buy, sell, and manage e-books efficiently. It helps users browse books, add them to the cart, place orders, and manage profiles online.

◆ 2. What technologies have you used in this project?

Answer:

We used **Java (JSP & Servlets)** for backend logic, **HTML, CSS, Bootstrap** for the frontend design, **MySQL** for the database, and **Apache Tomcat** as the server.

◆ 3. What is JSP and why did you use it?

Answer:

JSP (Java Server Pages) allows embedding Java code in HTML to create dynamic web pages. We used JSP for generating dynamic content like displaying books, handling forms, and showing order details.

◆ 4. What is a Servlet?

Answer:

A **Servlet** is a Java program that handles HTTP requests and responses in a web application. For example, `LoginServlet` handles login logic and redirects users accordingly.

◆ 5. How does your project connect to the database?

Answer:

We used **JDBC (Java Database Connectivity)** with a `DBconnect.java` class that establishes a connection to MySQL using:

```
Class.forName("com.mysql.cj.jdbc.Driver");
```

```
Connection conn = DriverManager.getConnection(url, username, password);
```

◆ 6. What is the role of DAO classes in your project?

Answer:

DAO (Data Access Object) classes, like `BooksDAOImpl` and `UserDAOImpl`, handle all database operations such as insert, delete, update, and retrieve data, separating logic from database code.

◆ **7. What tables have you created in your database?**

Answer:

We created:

- user – stores user details
 - book_dtls – stores book information
 - cart – stores cart items of users
 - book_order – stores order details
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◆ **8. What are the main modules of your project?**

Answer:

1. **User Module** (Login, Register, Cart, Order, Sell Old Books)
 2. **Admin Module** (Add, Edit, Delete Books, Manage Orders)
 3. **Database Module** (Handles storage and retrieval)
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◆ **9. How does the login system work?**

Answer:

The LoginServlet validates user credentials from the database using SQL.

If the credentials match, a **session** is created using:

```
session.setAttribute("userobj", user);
```

Otherwise, it redirects back to login.jsp with an error message.

◆ **10. What is Session Management in your project?**

Answer:

Session management keeps the user logged in until they log out. It uses HttpSession objects in servlets to store user data during browsing.

◆ **11. What is the purpose of MultipartConfig in your servlet?**

Answer:

It is used for file uploads — like when users add or sell a book and upload its image using:

```
@MultipartConfig
```

```
public class AddOldBook extends HttpServlet { ... }
```

◆ 12. How did you implement Add to Cart functionality?

Answer:

When the user clicks “**Add to Cart**”, book details are stored in the cart table using CartServlet and displayed in cart.jsp for checkout.

◆ 13. How does order placement work?

Answer:

When the user clicks “**Order Now**”, the system collects address and payment info, saves it into the book_order table, and clears the cart.

◆ 14. What payment method is available in your project?

Answer:

We implemented a **Cash on Delivery (COD)** option for simplicity. Future versions can include **Online Payment Gateway Integration**.

◆ 15. How does the admin panel work?

Answer:

The admin logs in using fixed credentials (like admin@gmail.com), then can **add new books, delete old books, and manage user records**.

◆ 16. What is the difference between RequestDispatcher and sendRedirect()?

Answer:

- RequestDispatcher forwards the same request internally (server-side).
 - sendRedirect() sends a new request to a different URL (client-side).
We used sendRedirect() after login to move to the home page.
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◆ 17. What are JSP directives?

Answer:

They provide global information to the JSP page, like:

```
<%@ page language="java" contentType="text/html; charset=UTF-8" %>
```

◆ 18. What are implicit objects in JSP?

Answer:

Objects like request, response, session, application, out, etc. — automatically available in JSP for handling client requests.

◆ **19. What are the key features of your project?**

Answer:

- User Login/Signup
- Buy and Sell Books
- Add to Cart and Checkout
- Profile Editing
- Admin Management
- Responsive Web Design

◆ **20. What is the flow of your project?**

Answer:

1. User logs in or registers
2. Views available books
3. Adds items to cart
4. Proceeds to checkout
5. Order confirmed and displayed in user order list

◆ **21. How is data security maintained in your project?**

Answer:

- Passwords are validated (and can be encrypted in real deployments).
- Session management prevents unauthorized access.
- Admin area access is restricted.

◆ **22. What challenges did you face while developing the project?**

Answer:

- File upload handling
 - Database connectivity errors
 - Session management bugs
 - Styling responsive pages using Bootstrap
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◆ **23. How did you handle database exceptions?**

Answer:

By using try-catch blocks and displaying user-friendly messages:

```
try {  
    // DB code  
} catch (Exception e) {  
    e.printStackTrace();  
}
```

◆ **24. How many JSP pages does your project contain?**

Answer:

Around **15–20 JSP pages**, including login.jsp, register.jsp, cart.jsp, order_success.jsp, edit_profile.jsp, sell_book.jsp, user_order.jsp, etc.

◆ **25. How did you implement Book Search functionality?**

Answer:

Using a search bar in navbar.jsp that passes the keyword (ch) to a servlet/JSP page and retrieves books using:

```
SELECT * FROM book_dtls WHERE bookName LIKE ? OR author LIKE ?;
```

◆ **26. What is the use of JSTL in JSP pages?**

Answer:

JSTL (<c:forEach>, <c:if>) simplifies looping and condition checking in JSP instead of writing Java code directly.

◆ **27. What tools did you use for development?**

Answer:

- **IDE:** Eclipse
 - **Server:** Apache Tomcat 9.0
 - **Database:** MySQL Workbench
 - **Version Control:** GitHub
 - **Frontend:** HTML, CSS, Bootstrap
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◆ **28. How do you deploy this project?**

Answer:

1. Export project as .war file from Eclipse
 2. Deploy in webapps folder of Apache Tomcat
 3. Start server and run via <http://localhost:8080/Ebook/>
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◆ **29. What is the future scope of your project?**

Answer:

- Online payment gateway integration
 - AI-based book recommendations
 - E-book reader integration
 - Admin analytics dashboard
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◆ **30. What did you learn from this project?**

Answer:

We learned full-stack web development using Java, database connectivity, servlet lifecycle, MVC architecture, team collaboration, and version control using GitHub.

💡 **Bonus Tip for Probe Presentation**

Each team member can prepare answers based on their module:

- **Vikas:** Backend (Servlets, DAO, DB connection)
- **Shrabanee:** Frontend (JSP UI, Bootstrap design)
- **Vasetti:** Admin features
- **Satyapriya:** User module (Cart & Orders)
- **Jyoti:** Database, schema, and ER diagram